

BROOKS, MI, USA

WMO: 725415

Lat: 42.251N

Lon: 84.956W

Elev: 941

StdP: 14.20

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54821

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.5	5.0	-8.1	3.7	1.2	-2.1	5.1	7.6	23.2	25.9	20.1	25.8	3.6	220	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	89.9	73.0	87.3	71.7	84.0	70.4	77.0	84.2	75.1	82.4	73.4	80.7	8.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.8	134.8	81.1	72.8	125.9	78.9	71.7	121.2	77.9	41.2	84.0	39.1	82.6	37.7	80.5	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	17.7	15.7	-10.2	93.3	6.6	2.8	-14.9	95.3	-18.8	96.9	-22.6	98.5	-27.4	100.6
			-10.5	79.9	6.3	2.4	-15.0	81.6	-18.7	83.0	-22.3	84.4	-26.9	86.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	24.1	25.9	36.1	48.2	59.4	68.4	72.1	70.2	63.8	51.5	40.6	29.6
	DBStd	19.21	11.35	11.14	11.40	9.53	8.72	6.72	5.56	5.37	7.72	8.76	9.51	10.02
	HDD50	3132	803	677	456	144	18	0	0	0	2	88	308	636
	HDD65	6422	1267	1095	899	509	219	40	8	15	114	427	731	1098
	CDD50	2873	1	2	24	90	310	551	685	627	417	135	28	3
	CDD65	685	0	0	1	5	46	141	228	177	78	9	0	0
	CDH74	6014	0	0	13	58	429	1234	2122	1406	677	74	1	0
	CDH80	1777	0	0	1	6	99	386	713	370	192	10	0	0
Wind	WSAvg	5.9	7.4	7.3	6.8	7.3	6.0	5.0	4.3	3.7	4.2	5.7	6.7	7.1
Precipitation	PrecAvg	36.2	2.4	1.8	2.1	2.9	4.0	4.1	3.7	3.7	3.4	3.2	2.7	2.2
	PrecMax	42.4	4.4	3.9	4.0	6.2	7.8	6.4	7.2	8.3	10.7	7.3	4.8	3.7
	PrecMin	28.7	0.6	0.5	0.6	0.3	1.7	1.5	0.8	1.1	1.2	0.4	1.1	0.9
	PrecStd	3.8	1.2	1.0	0.9	1.3	1.9	1.4	1.5	1.7	2.1	1.7	1.1	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.5	60.6	75.1	80.7	87.5	91.0	93.3	90.6	90.0	81.2	70.4	59.6
		MCWB	53.9	53.9	62.6	61.7	67.4	72.0	74.8	74.6	71.3	66.5	58.3	56.1
	2%	DB	49.5	54.0	65.8	74.6	82.3	88.2	90.0	87.6	85.8	75.2	63.8	53.8
		MCWB	46.0	49.3	57.2	59.4	66.4	70.9	74.4	72.6	71.0	63.3	55.2	50.9
	5%	DB	43.8	46.0	59.5	70.3	79.3	84.3	87.5	83.9	81.5	71.0	60.6	47.6
		MCWB	40.4	41.8	52.3	56.4	65.0	69.7	72.7	71.3	68.2	61.4	53.1	44.4
	10%	DB	38.7	40.4	54.2	65.5	74.8	81.3	83.9	81.4	77.9	66.3	55.4	42.6
		MCWB	36.4	36.6	47.7	54.3	63.4	68.3	71.1	69.9	66.8	59.2	49.1	39.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.7	56.2	62.8	66.6	72.7	76.2	80.8	78.5	76.3	70.6	61.5	56.6
		MCDB	55.9	57.9	73.8	75.8	79.8	84.8	85.9	87.2	84.2	77.8	66.7	59.3
	2%	WB	46.7	49.1	58.2	62.8	70.1	74.3	77.8	76.0	73.9	67.0	57.6	50.4
		MCDB	47.8	52.7	64.0	69.6	78.0	82.8	84.5	82.5	81.5	71.9	62.1	52.9
	5%	WB	40.8	42.2	53.5	59.1	67.9	72.7	75.6	74.1	71.4	63.3	54.3	44.7
		MCDB	43.5	46.3	58.7	66.6	75.8	81.0	83.2	80.4	77.3	67.7	59.3	46.5
	10%	WB	36.6	37.0	47.4	55.4	65.7	70.9	73.8	72.2	68.9	60.0	50.1	39.5
		MCDB	38.7	40.2	52.5	63.7	72.5	78.2	81.1	78.4	74.8	65.8	55.1	41.9
Mean Daily Temperature Range	5% DB	MDBR	13.4	15.2	17.7	20.9	20.3	19.6	20.6	19.8	20.8	18.5	15.3	11.9
		MCDBR	16.4	19.4	25.1	26.3	23.9	22.6	23.0	22.1	23.6	24.3	20.6	17.4
		MCWBR	14.2	15.7	16.9	14.7	12.4	9.8	10.0	10.4	11.1	13.5	14.3	15.0
	5% WB	MCDBR	15.3	18.6	22.4	22.4	20.3	18.6	19.1	17.7	19.1	19.4	18.8	16.3
		MCWBR	14.2	16.3	16.5	15.5	12.6	10.2	10.4	9.9	10.8	13.8	15.0	15.3
Clear-Sky Solar Irradiance	taub	0.298	0.358	0.348	0.391	0.424	0.443	0.430	0.434	0.388	0.349	0.317	0.302	
	taud	2.369	2.148	2.340	2.245	2.197	2.185	2.228	2.211	2.322	2.427	2.465	2.402	
	Ebn at Noon	268	262	284	278	271	265	267	262	267	266	259	255	
	Edh at Noon	26	38	35	42	45	46	43	43	36	29	23	23	
All-Sky Solar Radiation	RadAvg	454	731	1123	1461	1696	1916	1945	1676	1356	870	541	370	
	RadStd	43	100	112	135	161	124	91	84	123	87	74	41	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page